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Virtual flight test techniques to predict a blended-wing-body aircraft in-flight departure characteristics

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Abstract As one of the promising configurations of the next generation of commercial aircraft, research on departure characteristics of the Blended-Wing-Body (BWB) is of great signification to safe flight limits. A three-degree-of-freedom (3-DOF) virtual flight test in a wind tunnel has been implemented for a candidate configuration to predict the departure characteristics. The support mechanism, the test model and the control law of the virtual flight test are introduced. In order to show the relationship between virtual flight test and actual flight test, the similarity criterion is also given. In open loop, the model has mild oscillations in the longitudinal and lateral directions, which are stable in closed-loop. The effect of flight control has been verified in virtual flight and actual subscale flight test. The analysis of system identification results indicate that the model has a good response to the excitation signal, and the response is in reasonable agreement with the flight test. Finally, the virtual flight departure test results are compared with the flight test. It shows that there is a good correspondence between the angle of attack and the elevator deflection at departure. This gives promising evidence of the practicability of virtual flight testing to predict departure of a BWB.

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1. Introduction

With the continuous development of avionics and control technology, as well as the increasing requirements for aircraft fuel efficiency, carbon emissions and nitrogen oxide emissions, the Blended-Wing-Body (BWB) aircraft has become one of the viable options to replace traditional airliners because of its high aerodynamic performance.^{1–4} But it also faces various

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challenges, such as the need for multidisciplinary design optimization, trimming, stability and maneuverability.^{5–7} The stability and dynamic characteristics of the BWB aircraft are different from those of conventional aircraft. In particular, the lateral and directional stability and maneuverability places more demanding requirements on ensuring safety near the boundaries of the flight envelope of the BWB aircraft. The study of the complex flight state of the aircraft is of great significance in ensuring flight safety and prevention of flight accidents.^{8–10}

Studies on aircraft departure characteristics include wind tunnel tests, model free flight tests, and full-scale flight tests.¹¹ The conventional wind tunnel test method establishes a model through static and dynamic force test data to predict the maneuvering characteristics and flight quality of the aircraft.^{12–13} This method is effective at small angles of attack and low motion rates, but the results become increasingly uncertain at higher angles of attack and angular rates; prediction of behavior under extreme flight conditions via this approach may be inaccurate. Although actual flight tests can reflect real full-scale flight conditions very well, such tests are expensive and time consuming. In some cases, considering safety and other factors, the test flight envelope is also limited.^{14–15} In this context, Virtual Flight Test (VFT) technology in a wind tunnel emerges as a potentially powerful prediction/verification approach. This technology can fill the gap between conventional wind tunnel test and Flight Test (FT), enabling a reduction in the risk involved in flight testing, and can shorten the development cycle by reducing the number of trials.^{16–18}

Substantial research on aircraft flight dynamics and control has been carried out using virtual flight test technology. Pattinson et al. built a 5-DOF rig at University of Bristol (which can be operated in their 2.1 m × 1.5 m closed wind tunnel or the 1.1 m open-jet tunnel), and carried out multi-body dynamics modelling and simulation.¹⁹ Gatto et al. used a 2-DOF pendulum support and a 3-DOF mechanism in a 9 ft × 7 ft closed wind tunnel to test the M2370 dynamics similarity model, indicating that the technique works well in predicting longitudinal stability and control derivatives.²⁰ Carnduff et al. improved a 4-DOF experimental device and augmented the data measurement range and accuracy by Micro-Electro-Mechanism System (MEMS) in a 1.5 m × 1.1 m open wind tunnel, and conducted researches on the typical modal characteristics and parameter identification of aircrafts.²¹ In the wind tunnel named T-203 of Novosibirsk Academy of Aeronautics, Sohi applied a 3-DOF mechanism (Shtopor-203) to study spin and spin recovery of several kinds of fighters, including Su-26, Su-27 and Su-49.²² Khrabrov et al. recently developed a dorsal support dynamic institution to explore wing-rock characteristics of Mig-29 at high angle of attack, and verified control effects of modern robust control approach on wing-rock phenomenon through virtual flight test.²³

Through these studies, several key technologies were explored, including similarity criteria, support mechanisms, test models, and flight control systems.²⁴ It also shows that virtual flight test can be used for the study of aircraft departure characteristics. However, there is little discussion about the correlation between the experimental results of the virtual flight and the flight test. If the experimental results are consistent with the actual flight, the flight characteristics of the aircraft can be predicted by the virtual flight test to guide the design of the aircraft and the design of the control law. This

ensures that the aircraft avoids entering stalls, departure or even spin.

The paper is organized as follows. The Section 2 introduces the virtual flying system, including support mechanisms, models and control law. The Section 3 is about the flight test. The Sections 4 and 5 present the comparison of virtual flight test and flight test results, followed by the conclusions.

2. Virtual flight test description

The experiments were conducted in the low-speed wind tunnel of the College of Aerospace Engineering at Nanjing University of Aeronautics and Astronautics (NUAA), the dimensions of the test section of which are 2.5 m × 3 m. As shown in Fig. 1, the model is installed in the center of the wind tunnel through the support mechanism in the underside of the fuselage.

2.1. Wind tunnel model

The design of the BWB is based on previous works by Zhong²⁵ and Vicroy²⁶ et al. The planform incorporates three parts: the center-body, the trapezoidal wing and the outboard wing. The trapezoidal wings provide transition from the centre-body to the outboard wing. And airfoils are different in disparate sections as shown in Fig. 2.

The requirements for the model design are more complicated than traditional wind-tunnel tests. Not only must the aircraft model be geometrically similar, but also there are strict scaling requirements for the position of the Center of Gravity (COG), the weight, the inertia, and so on. Therefore, the model COG position can be adjusted, ensuring that it coincides with the center of rotation in the experiment. The three-axis inertia of the aircraft is measured by the cycloid method²⁷ and compared with the moment of inertia of the Computer Aided Tri-dimensional Interface Application (CATIA)²⁸ three-dimensional design model. The relative accuracy of each axial moment of inertia is greater than 90%, which meets the test requirements. The major parameters of the test model are provided in Table 1.

All sensor data and servo commands are communicated in 244 Hz in a wireless network between a central computer and three wireless nodes. All nodes are synchronized with the central computer by a simplified Network Time Protocol (NTP); maximum time delay is limited to within 100 ms. The time delay for message delivery is approximately 2 ms, and all messages attach their 64 bit time stamp in microseconds for

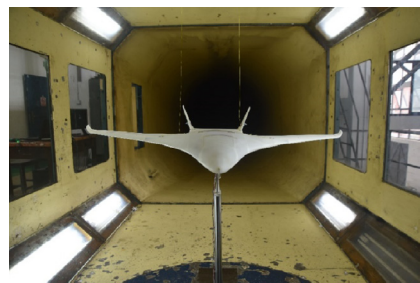


Fig. 1 Installation of BWB model for virtual flight test in wind tunnel.

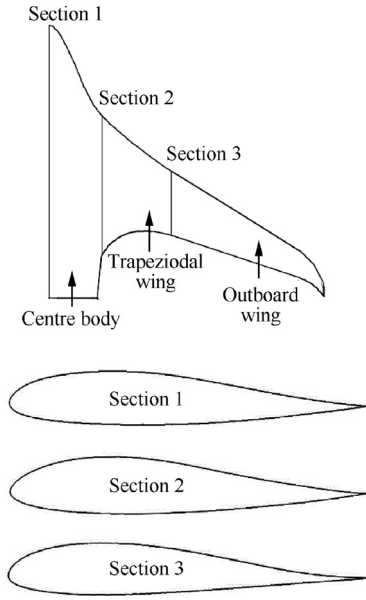


Fig. 2 Design of BWB: planform and airfoils for different sections.

Table 1 Geometric parameters of test model.

Parameter	Value
Reference area $S(\text{m}^2)$	0.250
Wing span $L(\text{m})$	1.54
Mean aerodynamic chord $c(\text{m})$	0.199
Mass $m(\text{kg})$	5.50
Rotational inertia of x -axis $I_x(\text{kg/m}^3)$	0.297
Rotational inertia of y -axis $I_y(\text{kg/m}^3)$	0.486
Rotational inertia of z -axis $I_z(\text{kg/m}^3)$	0.194

recording. On the central computer, an Extend Kalman Filter (EKF) is used to estimate absolute Euler angles of the aircraft model related to the wind tunnel and correct sensor biases. And there is a battery in the model to provide power.

The BWB model was manufactured with three-dimensional printing technology (Fig. 3), and incorporated motor servos for ailerons, elevators and rudder. Angle sensors are used to measure deflection angles. An embedded flight control system provides angular velocity and linear acceleration information for attitude calculation.

2.2. Support mechanism

Model support technology is the core of virtual flight test. It requires that not only the angular motion of the model around the center of gravity can be realized, but also the bearing capacity of the system can be satisfied, the interference of the support can be minimized, and the friction damping and structural vibration can be reduced.

It is a multi-bearing 3-DOF support mechanism (Fig. 4) that can allow each degree of freedom to be locked or released independently. The range of permitted rotation is 60° in pitch and roll, and 360° in yaw.

2.3. Control law

The aerodynamic data is obtained from static wind tunnel tests conducted in a 1 m low speed wind tunnel in NUA and the CFD simulation data provided by Shanghai Aircraft Design and Research Institute. The general linearized longitudinal differential equations of motion in flight dynamics can be expressed as follows.²⁹

$$\begin{bmatrix} \dot{\alpha} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -\frac{\bar{q} SC_{L\alpha}}{mU} & 1 - \frac{\bar{q} SC_{L(q+\dot{\alpha})}}{2mU^2} & 0 \\ \frac{\bar{q} SC_{m\alpha}}{I_y} & \frac{\bar{q} SC^2 C_{m(q+\dot{\alpha})}}{2UI_y} & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} -\frac{\bar{q} SC_{L\delta_e}}{mU} \\ \frac{\bar{q} SC_{m\delta_e}}{I_y} \\ 0 \end{bmatrix} \delta_e \quad (1)$$

where α is angle of attack, q is pitch rate, θ is pitch angle. $\dot{\alpha}$, $\dot{q}$ and $\dot{\theta}$ are derivatives with respect to α , q and θ respectively. $\bar{q}$ is the dynamic pressure, U means air speed, C_L is lift coefficient and C_m is pitch moment coefficient. $C_{L\alpha}$, $C_{L(q+\dot{\alpha})}$ and $C_{L\delta_e}$ are derivatives of C_L with respect to α , $q + \dot{\alpha}$ and δ_e respectively. $C_{m\alpha}$, $C_{m(q+\dot{\alpha})}$ and $C_{m\delta_e}$ mean derivatives of C_m with respect to α , $q + \dot{\alpha}$ and δ_e respectively.

A control law is developed using the aerodynamic model from which state-space models are extracted. For the purpose of stability augmentation and tracking demanded rates, we choose the Nonlinear Dynamic Inversion (NDI) method³⁰ with proportional-integral (PI) feedback control as the inner loop control frame, which tracks the angular rate commands $\mathbf{\Omega}_c = \{p_c, q_c, r_c\}$ and outputs $\{\dot{p}_c, \dot{q}_c, \dot{r}_c\}$ commands. The pseudo-linear controlled plan provided by NDI is

$$\dot{\mathbf{\Omega}} = \mathbf{0} \cdot \mathbf{\Omega} + \mathbf{I} \cdot \dot{\mathbf{\Omega}}_c = \dot{\mathbf{\Omega}}_c \quad (2)$$

where $\mathbf{\Omega}$ is the vector of angular rates and $\mathbf{I}$ is a unit matrix. The PI error controller is then designed based on this plan as shown in Fig. 5 (taking the longitudinal control law as an example).

The closed-loop transfer function in Fig. 5 is

$$\frac{q(s)}{q_c(s)} = \frac{K_i}{s^2 + K_p s + K_i} \quad (3)$$

where K_i and K_p are respectively the integral and proportional feedback gain coefficients. Comparing Eq. (3) with the standard second-order dynamics of

$$\frac{q(s)}{q_c(s)} = \frac{\omega^2}{s^2 + 2\zeta\omega s + \omega^2} \quad (4)$$

the closed-loop dynamics can be assigned directly by

$$K_i = \omega^2 \quad (5)$$

$$K_p = 2\zeta\omega \quad (6)$$

A suitable value of 12 rad/s is chosen for the natural frequency ω to increase the tracking performance, which is limited by the electronic speed regulator and servo delays. Herein, the damping ratio ζ is selected as 0.8 to satisfy the requirement of Level 1 at Stage A in USA Military Flying Qualities Specification (MIL-F-8785C). Attitude control can then be done directly as simple proportional control:

$$q_c = K_\theta(\theta_c - \theta) \quad (7)$$

where K_θ for time-scale separation.

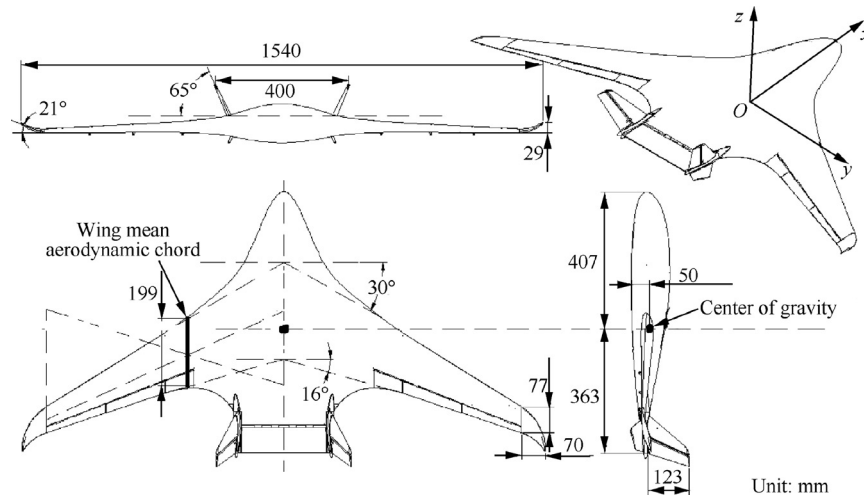


Fig. 3 BWB test model.

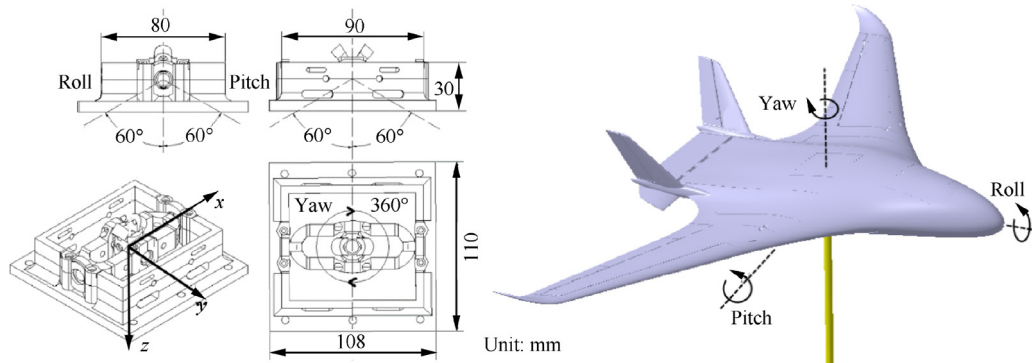


Fig. 4 Multi-bearing 3 DOF support mechanism.

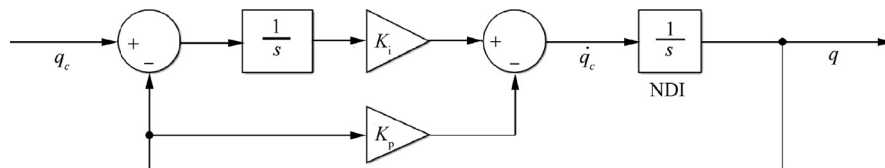


Fig. 5 Proportional-Integral (PI) error controller.

3. Flight test

For comparison, the flight and virtual flight test use the same control law. The precondition of validation for virtual flight in wind tunnel is similarity in dynamics. In a linear system view, it means keeping all matrix expressions the same after state conversion. Table 2 shows the similarity criteria used in this paper.

The main differences between the similarity criteria and conventional set³¹ are caused by choosing different time scaling parameter. The time scaling of conventional set is $K^{1/2}$, which means that all data from wind tunnel test is compressed by time compared with corresponding flight tests in air. Flight control system in wind tunnel has to sample sensors and calculate flight control law in higher frequency ($K^{1/2}$ times), which demands more powerful processor than real flight controller.

Table 2 Similarity criterion, where subscript aif refer to flight test aircraft.

Similar parameter	Expression	Ratio
Geometry length	L_{aif}/L	K
Geometry area	S_{aif}/S	K^2
Time	T_{aif}/T	1
Angle	α_{aif}/α	1
Speed	V_{aif}/V	1
Angular rate	$\dot{q}_{aif}/\dot{q}$	1
Angular acceleration	$\ddot{q}_{aif}/\ddot{q}$	1
Air density	ρ_{aif}/ρ	K_1
Moment of inertia	I_{xaif}/I_x	$K_1 K^5$
Mass	m_{aif}/m	$K_1 K^3$

Table 3 Geometric parameters of flight test aircraft.

Parameter	Value	Actual ratio	Theoretical ratio
Reference area $S(\text{m}^2)$	0.999	4.0	4.0
Wing span $L(\text{m})$	3.08	2.0	2.0
Mean aerodynamic chord $c(\text{m})$	0.39	2.0	2.0
Mass $m(\text{kg})$	20.8	3.9	5.2
Rotational inertia of x-axis $I_x(\text{kg/m}^3)$	4.54	15.3	20.9
Rotational inertia of y-axis $I_y(\text{kg/m}^3)$	6.46	13.3	20.9
Rotational inertia of z-axis $I_z(\text{kg/m}^3)$	2.99	15.4	20.9

Gyro sensor also need to support a larger range ($K^{1/2}$ times) for high agility maneuver simulation. And most critical demand is about actuators, which need higher ($K^{1/2}$) cut-off frequency to track deflection commands for control surfaces and $K^{1/2}$ times of maximum deflection rate.

On the contrary, new set of similarity conditions do not change time scaling, leading to same angular rate and acceleration for both motions and actuators between real and small-scale tests.

In Table 2, K_I is the correction parameter if moment of inertia between aircraft and model could not keep the ratio K^5 due to manufacturing reasons. However, from another view, it means the corresponding flight is at a different altitude rather than same altitude as the wind tunnel. In most cases, wind tunnels are built in low altitude area, thus K_I is less than one in most cases. Furthermore, the relationship of inertia

$$I_{\text{xaif}} = K_I K^5 I_x < K^5 I_x \quad (8)$$

means that a heavier model is acceptable, and better weight margins could be gained to balance moment of inertia in three axes.

The geometry ratio of the aircraft in flight test and the model in wind tunnel is $K = 2$. The parameters of the aircraft are in Table 3.

4. Test condition and procedure

Since the VFT is a 3-DOF rotation, and the FT is a 6-DOF motion, including three linear displacements and three angular motions, the effect of the linear displacement constraint between the VFT and the FT was carried out by simulation of longitudinal control response. The dynamic equation of 3-DOF motion simulation is as follows.

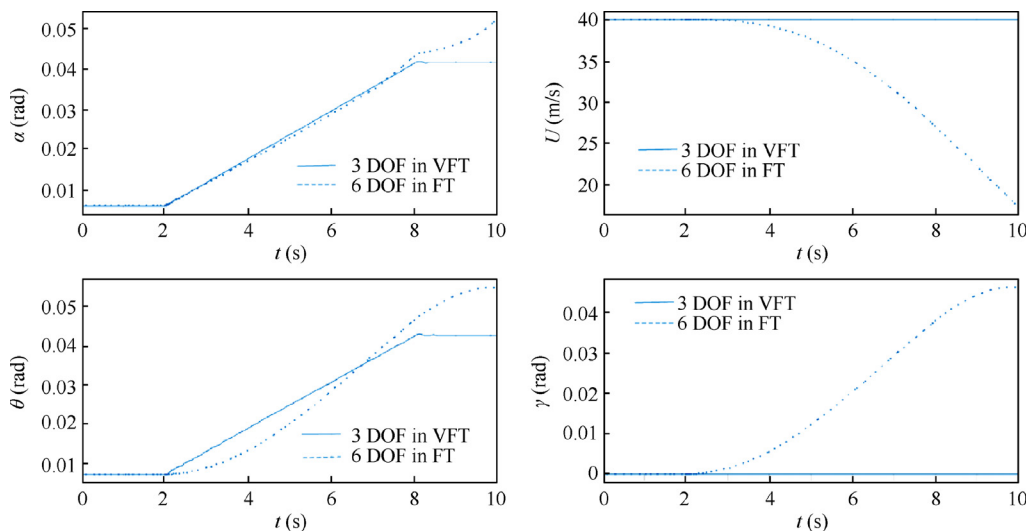
$$\begin{cases} I_x \dot{p} + (I_z - I_y)qr - I_{xz}(pq + \dot{r}) = L_m \\ I_y \dot{q} + (I_x - I_z)rp + I_{xz}(p^2 - r^2) = M_m \\ I_z \dot{r} + (I_y - I_x)pq + I_{xz}(qr - \dot{p}) = N_m \end{cases} \quad (9)$$

where L_m is the roll moment, M_m is the pitch moment and N_m is the yaw moment.

As shown in Fig. 6, The magnitude and direction of the wind speed remain unchanged in the VFT, and the linear displacement of the model is constrained, so the γ is always 0 rad. Compared with the FT, steady responses of α and θ decrease in the VFT. It is associated with the fast modes of motion, in which rotations play a much more important role in the short term than translational dynamics.

For the virtual flight test, the velocity is set to be 25 m/s, In the experiments, with all rotational degrees of freedom of the model free to be explored, the feedback control system maintained the roll/pitch/yaw angles of the BWB at approximately zero. Meanwhile, the deflection instructions sent from the ground station commanded the model. All the various signals (e.g. rotation speeds, tilt angles, elevon deflections) were recorded at a frequency of 200 Hz. The schematic of the VFT system is shown in Fig. 7.

There are three parts to the test. First, closed-loop stability test. The effect of the flight control system is verified by comparing the open-loop and closed-loop attitude stability. Second, identification of dynamic derivatives. Given a standard excitation signal, the corresponding response time history of the model is obtained. Then the dynamic derivatives are obtained by parameter identification. Finally, departure test. In closed-loop, by slowly changing the elevator input, the angle of attack increases gradually until departure occurs.

**Fig. 6** Comparison of longitudinal control response between VFT and FT.

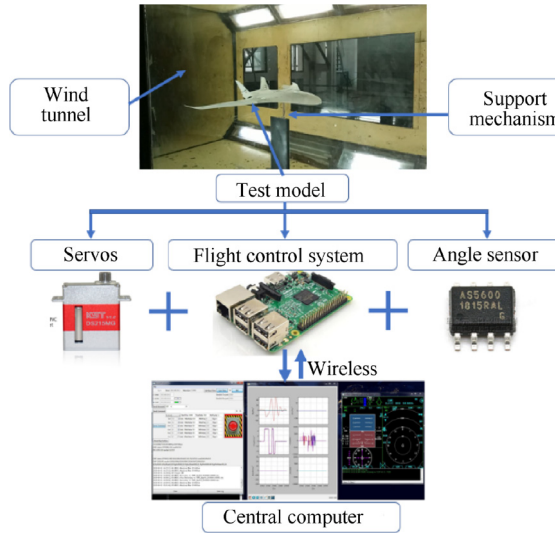


Fig. 7 Virtual flight test system.

The flight test also includes the above test conditions. Neither of these two tests can directly obtain the flow angle, so it is necessary to calculate the flow angle through the attitude angle. In wind tunnel, the flow is consistent with the axis of the wind tunnel, so the Earth-axis system X_d is consistent with the wind axis system X_q . Through coordinate conversion:

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \mathbf{B}_t^d \mathbf{B}_q^t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (10)$$

where $\mathbf{B}_t^d$ is the transformation matrix from the body axis system to the earth axis system, and $\mathbf{B}_q^t$ is the transformation matrix from the wind axis system to the body axis system.

The angle of attack α and sideslip angle β can be obtained by equation

$$\tan \alpha = \frac{\cos \varphi \sin \theta \cos \psi + \sin \varphi \cos \psi}{\cos \theta \cos \psi} \quad (11)$$

$$\sin \beta = \sin \theta \sin \varphi \cos \psi - \sin \psi \cos \varphi \quad (12)$$

where θ is the pitch angle, φ is the roll angle and ψ is the yaw angle.

For flight test, the Polaris Autopilot developed by Zerotech was used for this paper. It supports both manual Remote Controlled (RC) mode and autonomous mode. The airborne avionics includes a gyroscope, an accelerometer, a Global Positioning System (GPS) and a data modem. Detailed specifications of the Polaris Autopilot are shown in Table 4.

Attitude angles and angular rates information can be obtained, and the extended Kalman filter can ensure that the sensor with low Signal-To-Noise Ratio (SNR) can output atti-

tude information with high truth value. All the sensor data is logged onboard the aircraft including inertial data (100 Hz), GPS data (10 Hz). The airflow angle α can be obtained by equation:

$$\alpha = \theta - \gamma \quad (13)$$

5. Virtual flight test and flight test comparisons

5.1. Closed-loop stability test

Fig. 8 shows the pitch angle and roll angle in VFT and FT. For VFT, the attitude information is recorded in the open-loop (input value: $\delta_e = 0^\circ$, $\delta_r = 0^\circ$, $\delta_a = 0^\circ$) and closed-loop. For safety reasons, the flight test is only conducted in the closed-loop condition.

According to Fig. 8, in the open-loop state, the BWB basically maintains a trim angle of attack of about 0° with slight oscillation of 2° . The roll angle also has a oscillation of 2° . When the loop is closed, the attitude angle (pitch and roll) can maintain at 0° and almost no oscillation. Therefore, this result indicated that the control law verified in VFT can also be used in FT. It also lays a foundation for the prediction of departure characteristics by VFT.

5.2. Identification of aerodynamic derivatives

Next, the correlation between VFT and FT will be further verified through the identification of aerodynamic derivatives. The identification process is shown in Fig. 9.

First, the attitude of the aircraft is kept stable by closed loop. Then, a doublet signal is sent by ground station to make the aircraft response. At this time, the aircraft is also switched from closed-loop to open-loop state. When the excitation action is over, the aircraft returns to the closed loop. All sensor data, servo commands and attitude information are recorded. The Output Error Method (OEM) based on maximum likelihood method is used to identify the unknown parameters. Optimization function $J(\Theta) = \det(\mathbf{R})$ and its derivative expansion $\frac{\partial J}{\partial \Theta} = \mathbf{G}$ are as follows

$$\mathbf{R} = \frac{1}{N} \sum_{k=1}^N [\mathbf{z}(t_k) - \mathbf{y}(t_k)][\mathbf{z}(t_k) - \mathbf{y}(t_k)]^T \quad (14)$$

$$\mathbf{G} = - \sum_{k=1}^N \left[\frac{\partial \mathbf{y}(t_k)}{\partial \Theta} \right]^T \mathbf{R}^{-1} [\mathbf{z}(t_k) - \mathbf{y}(t_k)] \quad (15)$$

where $\mathbf{y}$ is the pitch angle and pitch angle rate from the calculation model, and $\mathbf{z}$ is the corresponding pair of values obtained from the experiment. t_k is the sampling time, and N is the number of maneuver sample. Taking a longitudinal

Table 4 Specifications for the Polaris Autopilot.

Parameter	Measurement range	Non-linearity	Initial bias error	Noisy density
Gyroscope	300 ($^\circ$)/s	$\pm 0.03\%$ fs	$\pm 0.25(^\circ)$ /s	$0.03/\sqrt{\text{Hz}}$
Accelerometer	± 5 g	$\pm 0.1\%$ fs	± 0.0002 g	$4 \times 10^{-5}/\sqrt{\text{Hz}}$

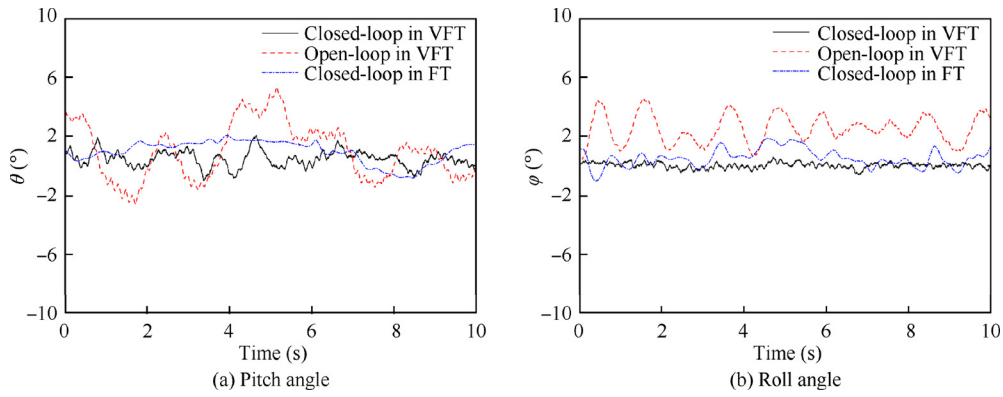


Fig. 8 Attitude angles.

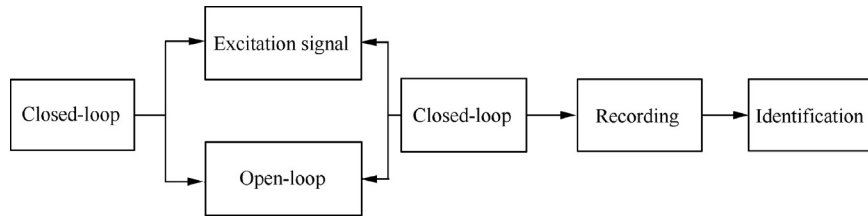


Fig. 9 Identification process.

example, the longitudinal dynamic response of VFT and FT is contained in Fig. 10.

It shows that, the trimmed elevator is similar between VFT and FT. The tendency of angle of attack α and pitch angle rate q is consistent but the peak values are different. They have a basically the same α and q has a 0.15 rad/s deviation. Table 5 shows the comparison between identification results and dynamic derivatives obtained from computational fluid dynamics simulation. For VFT and FT, identification results are similar. The relative error of $C_{m(q+\dot{z})}$ is 5.1%, and the error of C_{mz} is 11.7%. Considering that the following departure test is a slow process, such difference is acceptable and does not affect the corresponding relationship between virtual flight test and flight test. Compared with the results of traditional identification approach (CFD), the relative error of $C_{m(q+\dot{z})}$ is 18.8%, and the error of C_{mz} is 20.0%. CFD is a complete numerical method, it depends on the accuracy of mathematical models. And considering that VFT and FT are more realistic simulation of motions, the relative error is reasonable.

5.3. Departure test

The departure is in the closed-loop situation. In the case of maintaining the stability of the aircraft, slowly increase the elevator input, so that the aircraft continues to pitch-up until attitude divergence occurs.

Fig. 11 shows attitude angles and angle rates obtained in VFT. The α first increases slowly as the δ_e decreases, and roll angle ϕ and yaw angle ψ keep near 0° within this time. When $t = 51.7$ s the δ_e is -0.22 rad, and the corresponding angle of attack is 0.22 rad. Then the elevator starts to increase in reverse, which indicates that the aircraft is longitudinally unstable. However, because the aircraft is in a closed loop state, it is not immediately unstable. The α still increases to

0.24 rad stably at $t = 52.5$ s. But at the same time the roll rate and the yaw rate decrease to -0.4 rad and -0.3 rad respectively, which make the roll angle reduce to -0.15 rad and yaw angle to -0.1 rad. Due to the coupling of the lateral and directional dynamics, the angle of attack rises rapidly to 0.33 rad at $t = 53.0$ s. After that, the flight controller can no longer maintain the stability of the aircraft. The BWB finally enters a three-axis oscillation. This can also be considered as departure or loss-of-control. And the critical angle of attack where departure occurs can be considered as 0.24 rad.

In our early research of the BWB, force measurements and flow field analyses were included.³² The results show that the BWB has a good lift efficiency. Its stall angle is 0.6 rad. But the flow separation at the outboard makes the longitudinal characteristics change at a low angle of attack about 0.17 rad. And due to the large wingspan, the asymmetric flow separation that occurs on the outer wing will cause roll instability (Fig. 12). Through the comparison with the previous work, the departure characteristics of the aircraft can be understood.

The same process is also carried out in the FT to compare with the VFT. For aircraft safety, a stall recovery criterion is established based on the results of the VFT. The stall criterion is shown in Table 6. When any condition is triggered, the aircraft will recover out of stall.

Fig. 13 is the time history of attitude angles and angle rates. The δ_e is generally consistent with the trend of virtual flight. It also has a reversal. And at the inversion point, the α is 0.24 rad corresponding to -0.25 rad of elevator when time is 1641.8 s. After the reversal of δ_e , the roll rate and the yaw rate began to increase like the phenomenon of the VFT. Because the flight test has stall recovery measures, when it reaches the stall condition, it will automatically return the aircraft to the trim state. As shown in Fig. 10, What triggered the stall

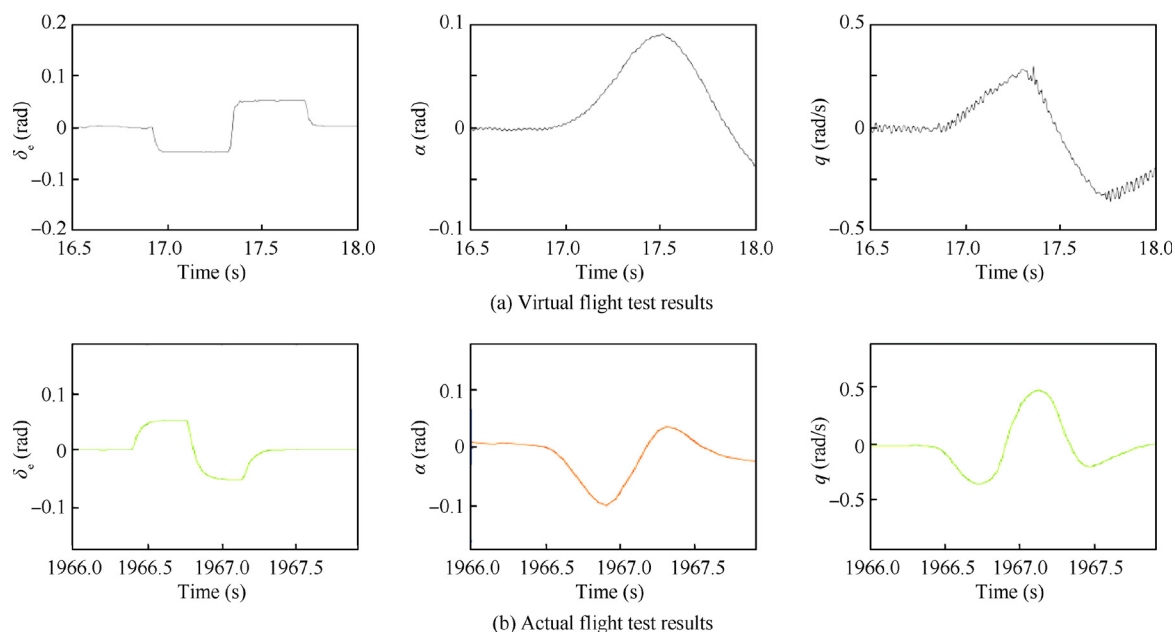


Fig. 10 Longitudinal dynamic response.

Table 5 Identification results of VFT and FT.

Parameter	VFT	FT	CFD	Relative error (VFT and FT)	Relative error (VFT and CFD)
Static derivative $C_{m\dot{\alpha}}$	-0.60	-0.53	-0.72	11.7%	20.0%
Dynamic derivative $C_{m(q+\dot{\alpha})}$	-8.50	-8.93	-6.9	5.1%	18.8%

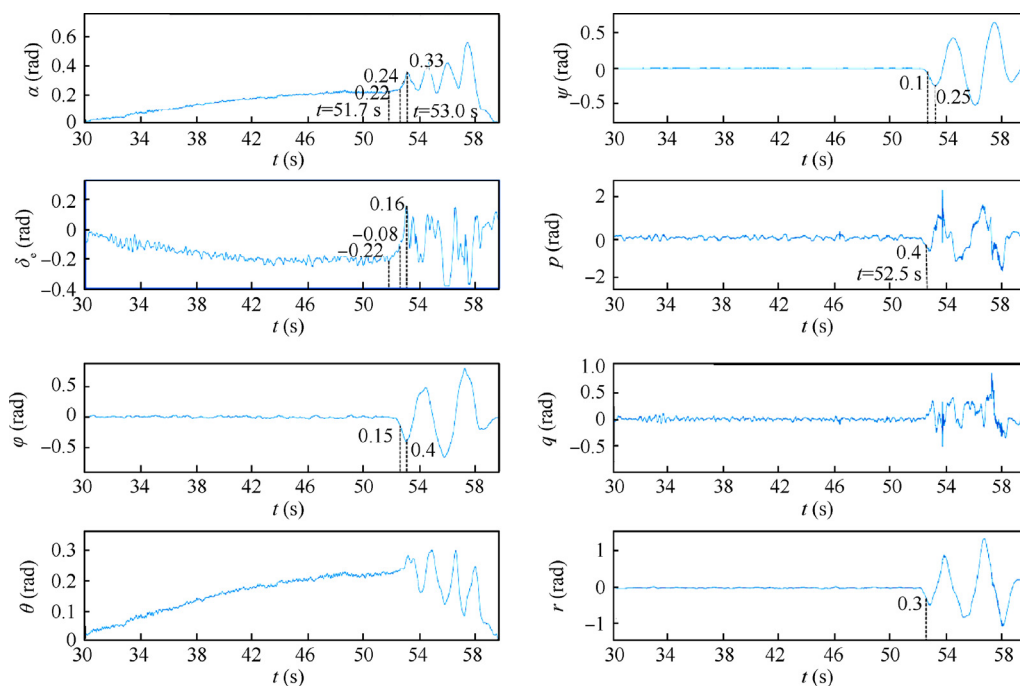


Fig. 11 Departure test results in VFT.

recovery was that the yaw rate exceeded 0.2 rad at $t = 1642.3$. In this time, the angle of attack is 0.26 rad, which can be as a sign of departure.

The comparison of flight test and virtual flight test results shows that the trends of angle of attack is almost the same, and the elevator input is also basically equal. It also shows that

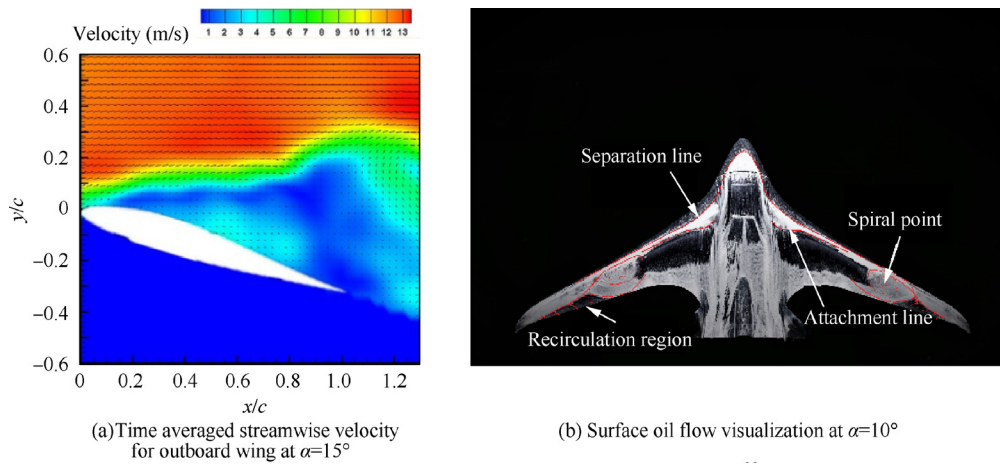


Fig. 12 Flow field analyses from previous work.³²

Table 6 Stall criterion of flight test aircraft.

Parameter	Value
Roll rate $ p $ (rad/s)	≥ 0.4
Pitch rate $ q $ (rad/s)	≥ 0.3
Yaw rate $ r $ (rad/s)	≥ 0.2
Pitch angle $ \theta $ (rad)	≥ 0.4
Roll angle $ \phi $ (rad)	≥ 0.3

VFT and FT have the same static stability region and their onset angle of departure is almost the same. So, the virtual flight test predicted the departure characteristics of BWB during flight test.

6. Conclusions

- (1) A wind tunnel virtual flight test system consisting of a support mechanism, test model, flight control and ground control system was developed. The model's mass, the center of gravity and moment of inertia design follow the dynamic similarity criteria relative to a larger flight test model. Through this design, the similarity between the virtual flight test and flight test is satisfied to ensure the reliability of the virtual flight result.
- (2) The effect of stability augmentation and attitude control is verified in the virtual flight test. And a new set of similarity conditions do not change time scaling, leading to same angular rate and acceleration for both motions and actuators between real and small-scaled tests. In the

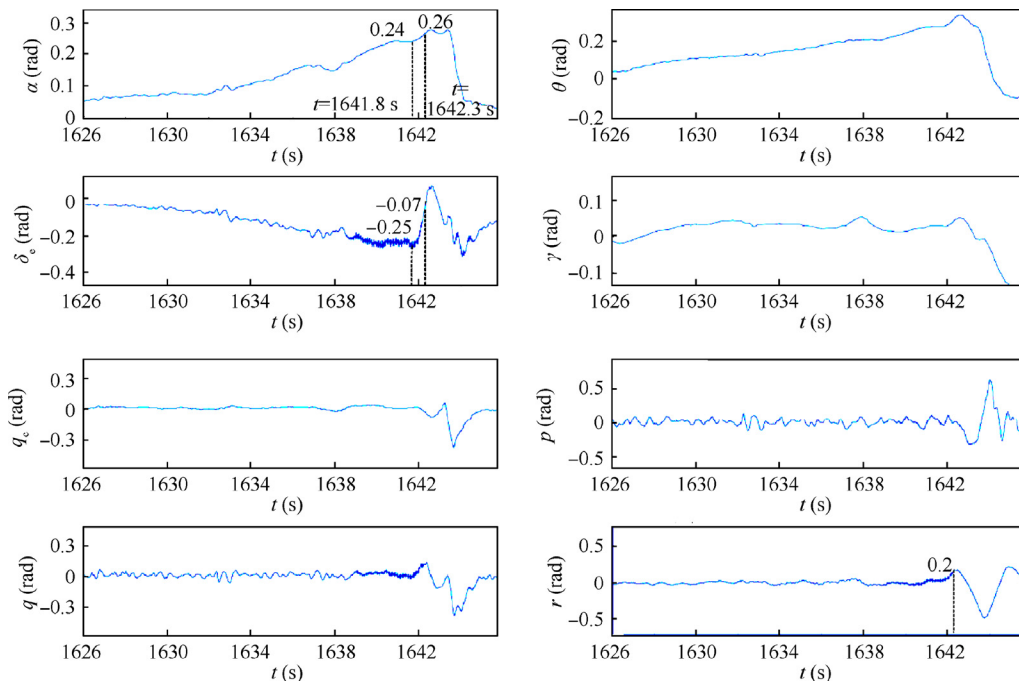


Fig. 13 Departure test results in FT.

longitudinal identification test, the pitch angle speed and angle of attack response processes are basically consistent with the flight test results, and the difference in angle of attack response is within 0.06 rad.

- (3) Through the VFT, departure characteristics of the BWB are presented in physics experiments. The support mechanism and freely moving model with flight control allows deeper insight into departure conditions. The obtained test results clearly indicate the process of departure. At the critical angle of attack, the divergent motion of lateral and directional course makes it difficult to maintain stability in the longitudinal plane, resulting in the divergence of the entire aircraft flight attitude. Based on the results of the VFT, reasonable stall boundary conditions were designed to ensure the safety of the FT.
- (4) Due to the understanding of departure characteristics of the BWB from the VFT, the FT was successful. Comparing with the VFT, FT produces the same static stability region and the onset angle of departure is almost the same. And the stall criterion prevented the BWB from loss-of-control. It shows that the virtual flight test can predict the departure characteristics of the flight test BWB during flight test.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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